

Image Transformation

Short quiz

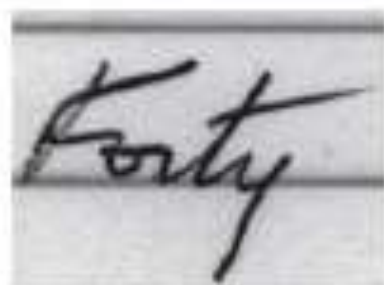
- This is MRI image of brain
 - Target is tumor detection
(It can be seen at the center)
 - Problem: Data size reduction is necessary before process
 - Question: Reduce resolution or bit depth or any other suggestion?
-
- Notice: There is not only one correct answer (as well as there is not only one wrong answer!)



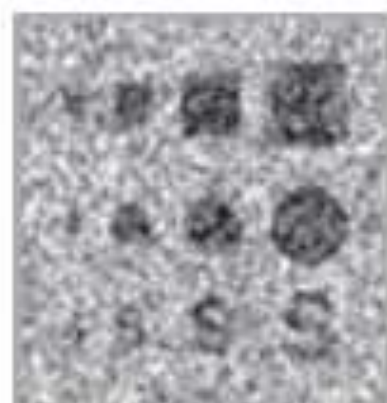


Image enhancement

- Improve the image properties for specific application

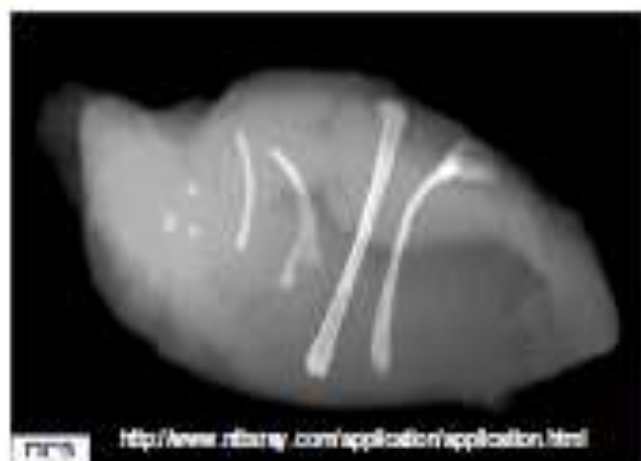
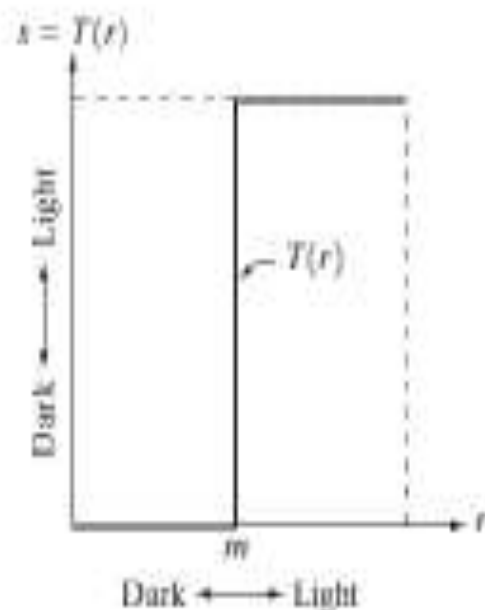


- Two main categories:
 - Spatial
 - Point operation
 - Local operations
 - Frequency

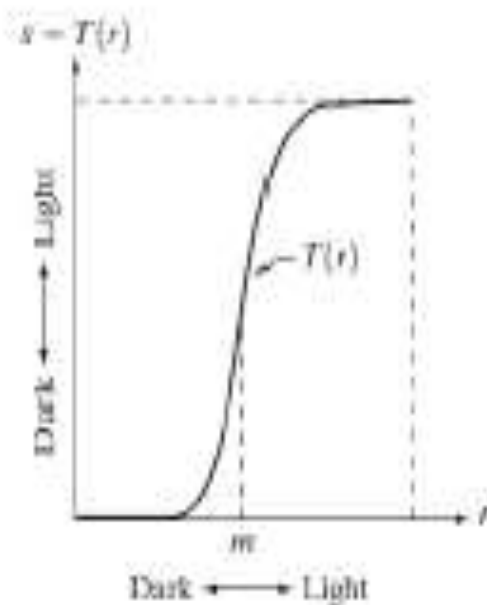


Thresholding

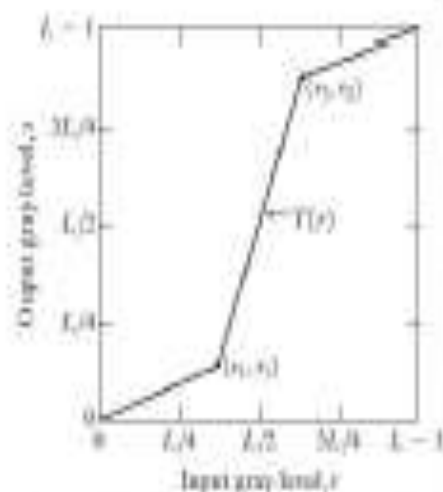
- Example in food inspection



Contrast stretching



Contrast stretching



u
v
w
x
y
z

FIGURE 8.10

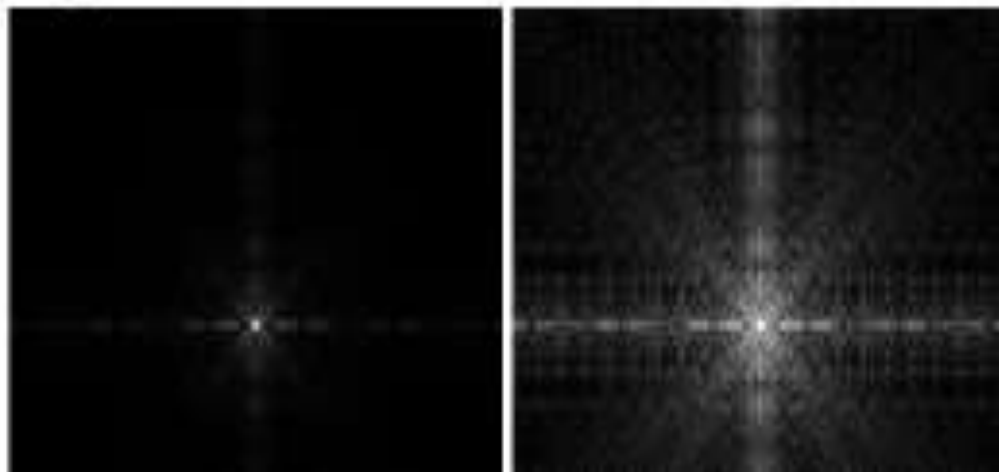
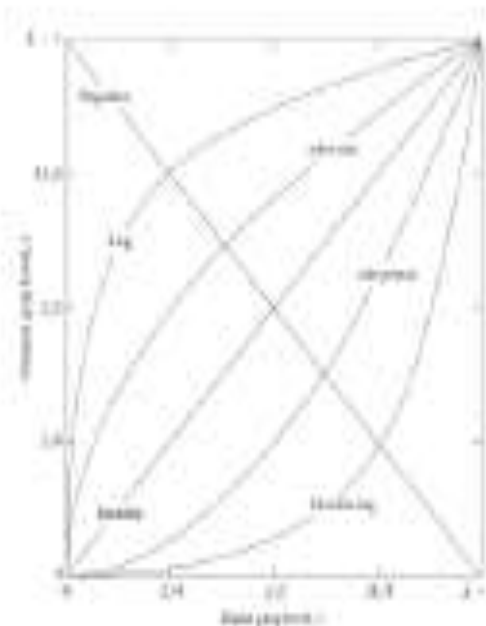
Contrast stretching. (a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Hoody, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

Log transform

- Expand the dark dynamic range

$$s = c \log(1 + r)$$

- Inverse log will do the opposite way

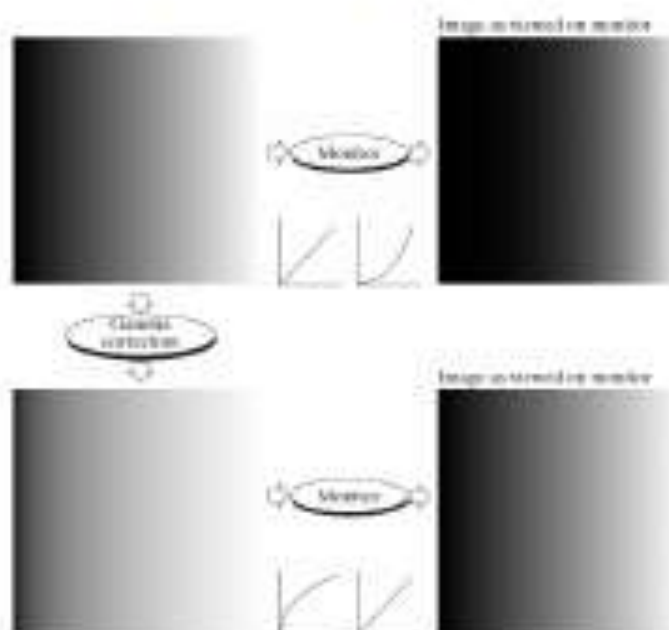
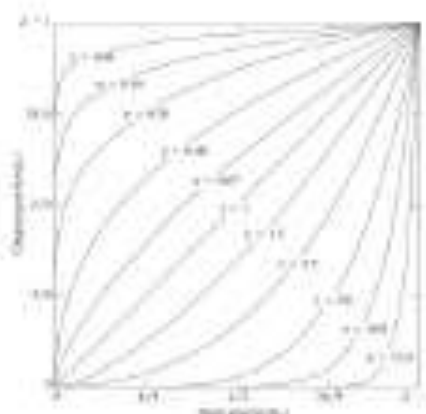




Power-law transform

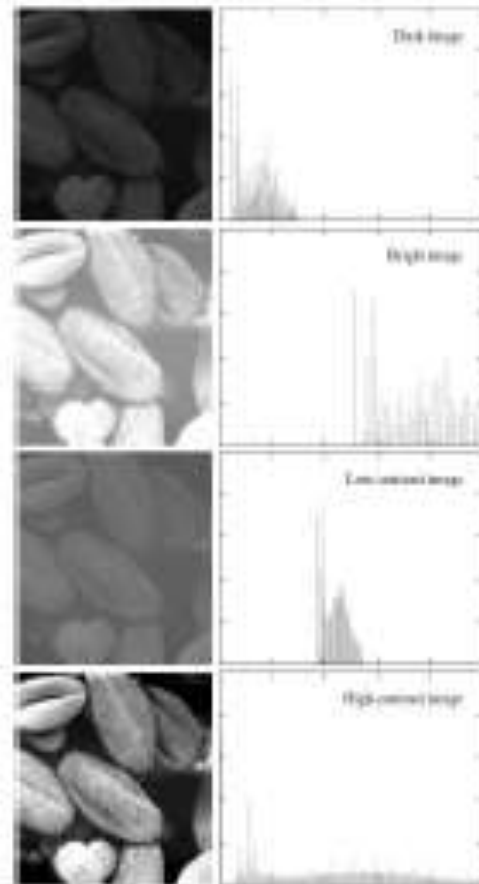
- Is similar to Log transform
- Referred as Gamma transform too because of its notation

$$s = cr^\gamma$$
- Many of imaging devices have power-law response and need Gamma correction



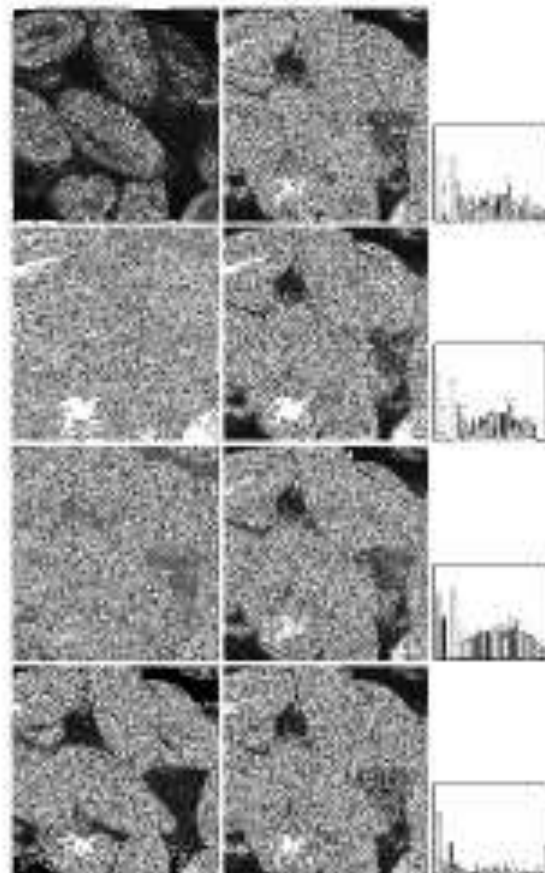
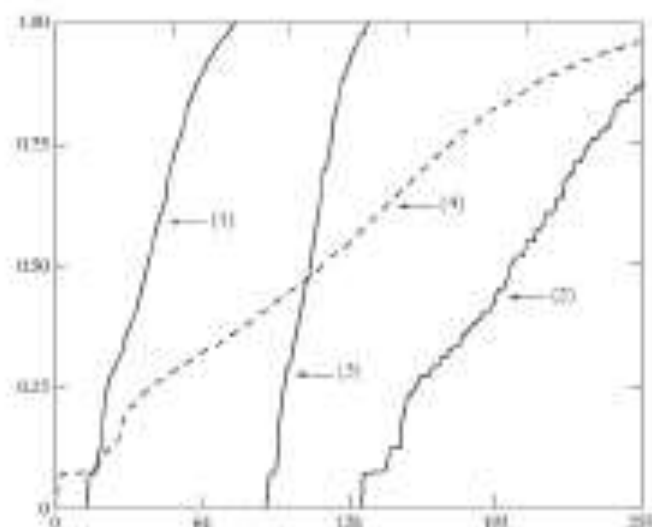
Histogram

- $h(L)$ = number of pixels with gray level L

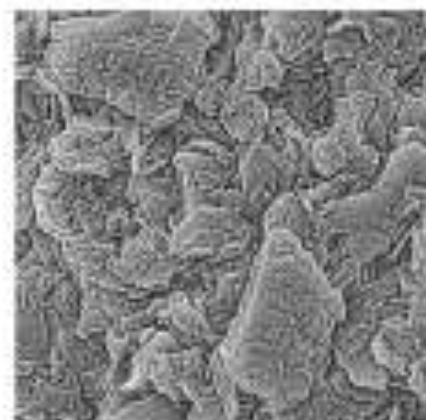
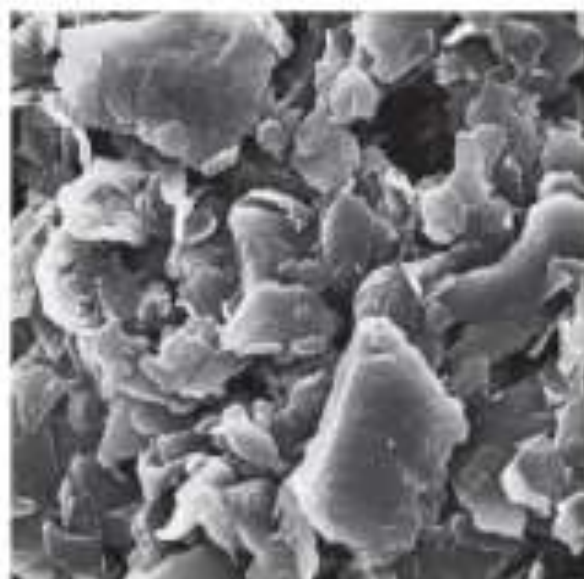


Histogram equalization

$$\begin{aligned}s_k &= T(r_k) = \sum_{j=0}^k p_r(r_j) \\ &= \sum_{j=0}^k \frac{n_j}{n} \quad k = 0, 1, 2, \dots, L-1.\end{aligned}$$



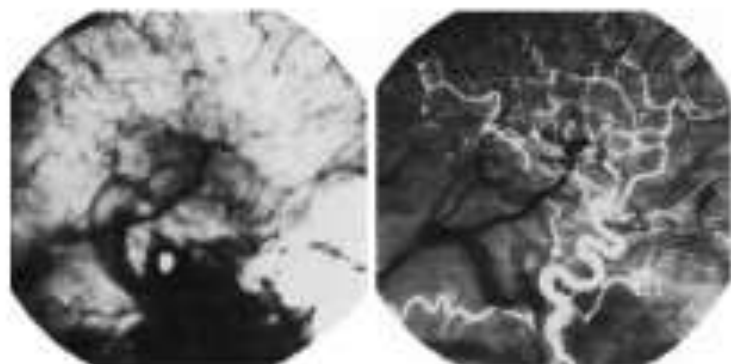
Local histogram equalization



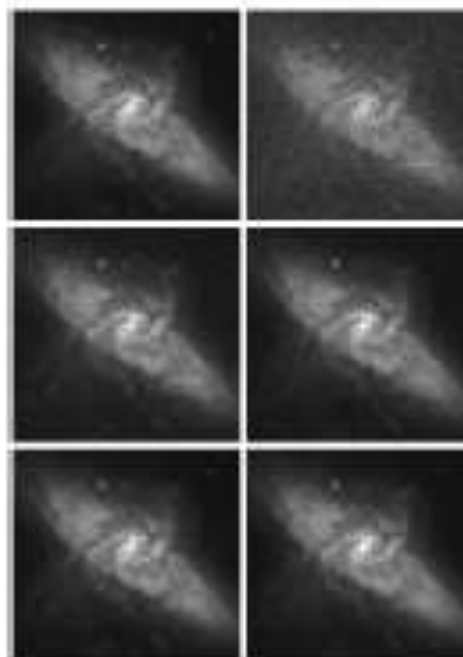


Arithmetic calculation

- Add, Subtraction, Average, AND, OR, ...



Subtraction:
Mask mode radiography



Averaging: Noise removal
in astronomy

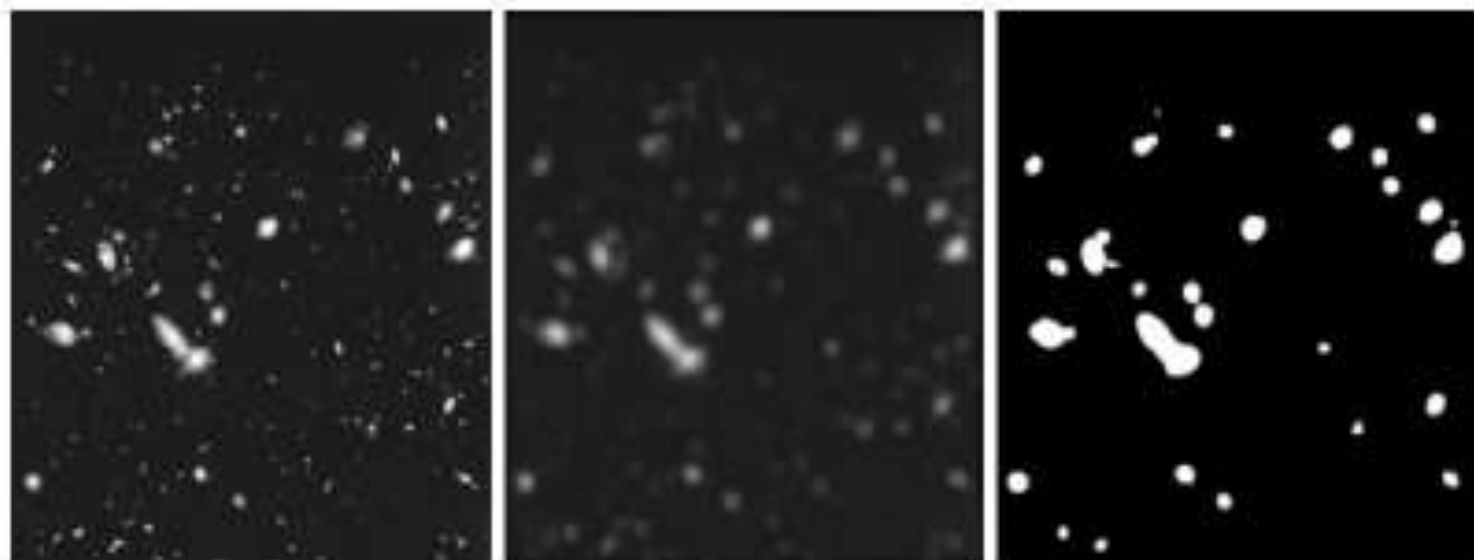


Smoothing linear filter

- Average / low pass filters
- Usually **noise** is unwanted sharp transitions in gray level
- By local averaging we **reduce sharp** transitions (irrelevant details)
- The side effect would be losing desired sharp transitions like **edges**
- Example of smoothing filters:
Box filter, and Weighted average filter

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array} \quad \frac{1}{16} \times \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

Average filter example



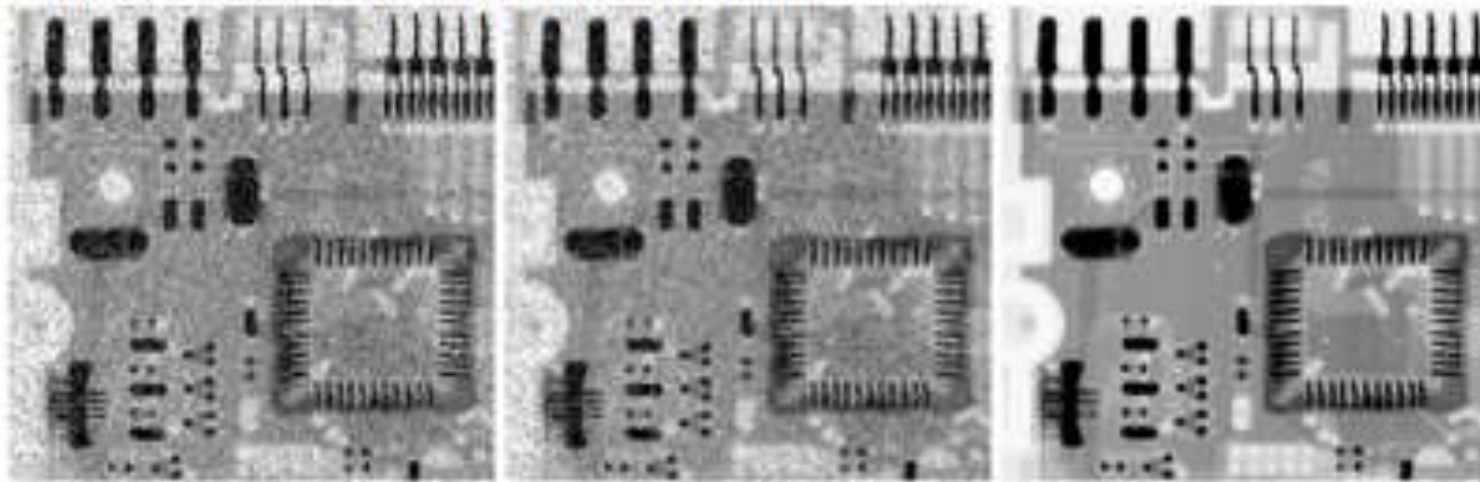
a b c

FIGURE 3.34 (a) Image from the Hubble Space Telescope. (b) Image processed by a 15×15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)



Order-statistic filters

- Median, min, max, and similar filters
- Select the result value based on one statistical property of the gray level distribution of pixels under the mask
- Good for removing some known type of noises like *impulse* or *salt-and-pepper*
- Example of reducing salt-and-pepper noise by Mean and Median filters respectively from left to right





Sharpening spatial filters

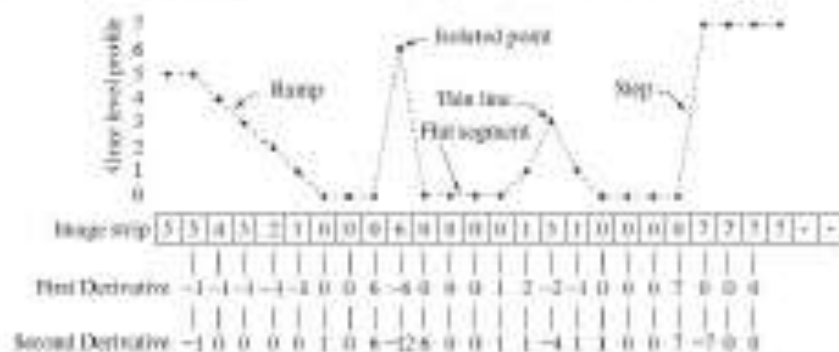
- Edge / High-pass filters
- Enhance the **details** of the image
- Improve sharp transitions
- The side effect can be increasing undesired details like **noise**
- The basic idea is that sharp transitions cause good response in the first or second **derivative**

$$\frac{\partial f}{\partial x} = f(x+1) - f(x),$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x).$$



Effect of image derivative





Some derivative masks (Laplacian)

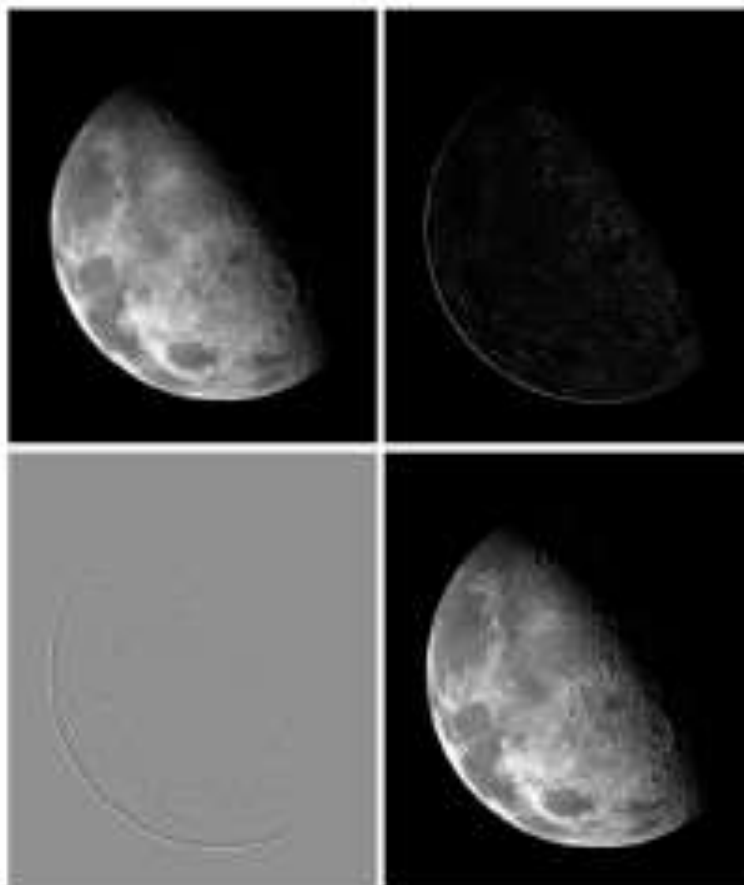
0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1
0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

Sharpening by Laplacian



(a) (b)
(c) (d)

FIGURE 3.48
(a) Image of the North Pole of the moon.
(b) Laplacian-blurred image.
(c) Laplacian image scaled for display purposes.
(d) Image enhanced by using Eq. (5.3.5).
(Original image courtesy of NASA.)

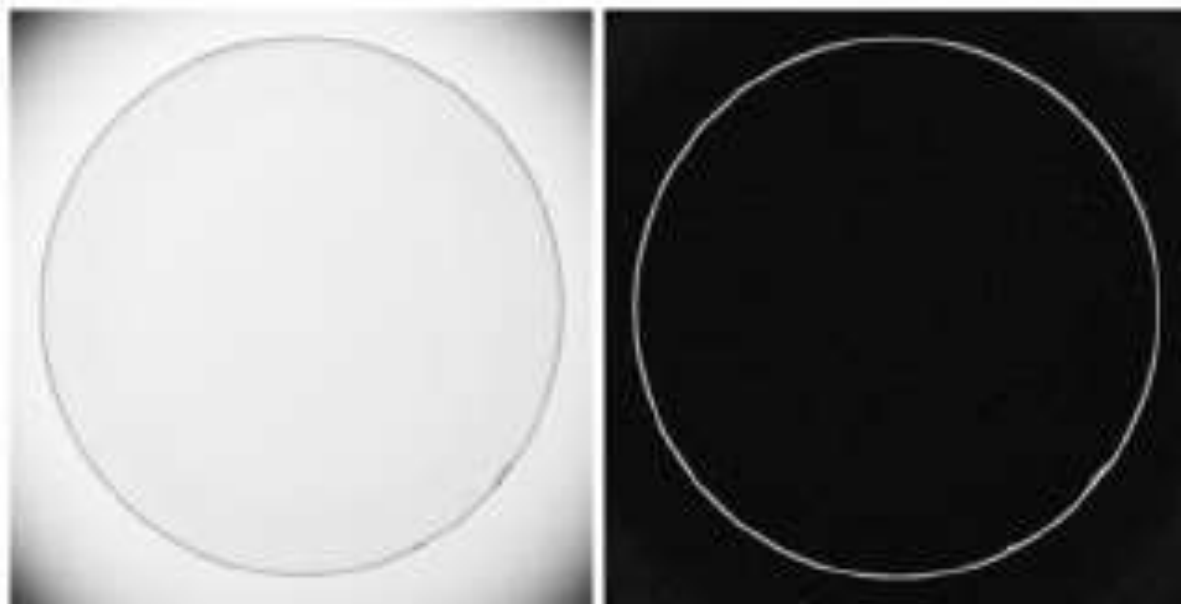


First derivative as edge detector (Gradient)



Sobel filter:

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	1	0	-1



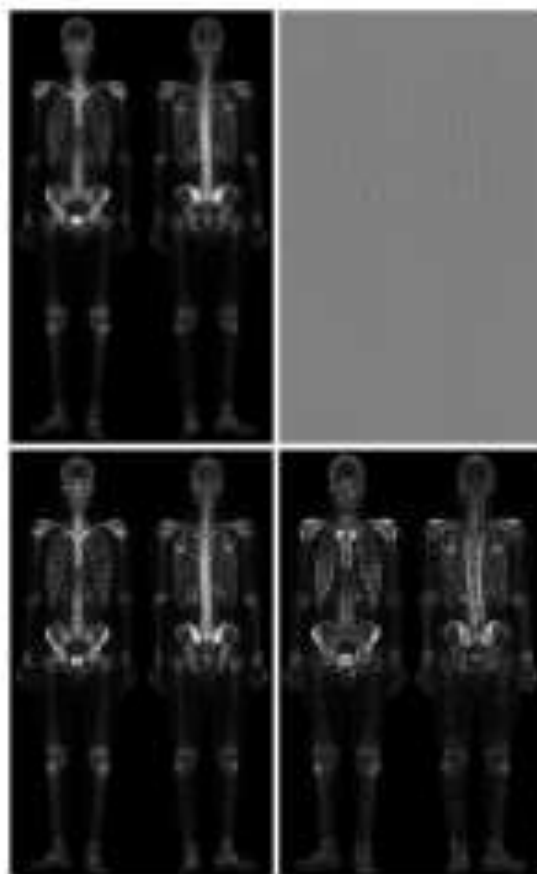
Combination of filters



0.00
0.00

FIGURE 3.44

(a) Images of whole body bone scan. (b) Spectrum of (a). (c) Stopped image obtained by adding to (a) the spectrum of (b).



0.00
0.00

FIGURE 3.45

(a) Commonly provided image showing ratio of 2 X 2 area (top) like (b) and (c) image formed by the product of (a) and (b). (d) Stopped image obtained by the sum of (a) and (b). (e) Final image obtained by applying a Gaussian filter to (d). (f) Composite image of (a) and (b) with ratio of (a) and (b) image.

